

2022-Ph-H1-Q19

Section: Particles and Waves

Topic: Wave-Particle Duality

What does increasing the frequency of a photon do?

Answer: E

(Increases the energy of the photon)

Guidance for Students:

Photon energy is given by $E = hf$. Higher frequency means higher energy, and shorter wavelength.

Revision Tips:

- $E = hf$, where h is Planck's constant.
- Frequency $\uparrow \rightarrow$ energy $\uparrow \rightarrow$ wavelength $\downarrow$
- Applies to all forms of electromagnetic radiation.